MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, LOW-POWER SCHOTTKY TTL, MAGNITUDE COMPARATORS, MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, low-power Schottky TTL, magnitude comparator microcircuits. Two product assurance classes and a choice of case outlines and lead finishes are provided and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510, and as specified herein.

1.2.1 *Device type.* The device type shall be as follows:

Device type	Circuit
01	4-bit comparator

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Outline letter	Case outline (see MIL-M-38510, appendix C)
E	D-2 (16-lead, 1/4" x 7/8"), dual-in-line package
F	F-5 (16-lead, 1/4" x 3/8"), flat package
2	C-2 (20-terminal, .350" x .350"), square chip carrier package

1.3 Absolute maximum ratings.

Supply voltage range	-0.5 V dc to +7.0 V dc
Input voltage range	-1.5 V dc at -18 mA to +5.5 V dc
Storage temperature range	-65°C to +150°C
Maximum power dissipation (P_D) 1/	110 mW dc
Lead temperature (soldering, 10 seconds)	+300°C
Thermal resistance, junction-to-case (θ_{JC}):	
Case E	50°C/W
Case F	70°C/W
Case 2	60°C/W
Junction temperature (T_J)	+175°C

1/ Must withstand the added P_D due to short circuit test (e.g., I_{OS}).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) *appearing at the end of this document or by letter.*

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1.4 Recommended operating conditions.

Supply voltage (V_{CC})	4.5 V dc minimum to 5.5 V dc maximum
Minimum high-level input voltage (V_{IH})	2.0 V dc
Maximum low-level input voltage (V_{IL})	0.7 V dc
Normalized fanout (each output)	10 maximum
Case operating temperature range (T_C)	-55°C to +125°C

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Terminal connections and logic diagrams. The terminal connections and logic diagrams shall be as specified on figures 1 and 3, respectively.

3.2.2 Truth tables. The truth tables shall be as specified on figure 2.

3.2.3 Schematic circuits. The schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in this specification and shall be submitted to the qualifying activity as a prerequisite for qualification. All qualified manufacturers' schematics shall be maintained and available upon request.

3.2.4 Case outlines. The case outlines shall be as specified in 1.2.3.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 and 6.4 herein.

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range, unless otherwise specified.

3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.6 Marking. Marking shall be in accordance with MIL-M-38510.

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions $-55°C \leq T_C \leq +125°C$	Limits		Unit
			Min	Max	
High-level output voltage	V_{OH}	V_{CC} = 4.5 V, V_{IH} = 2.0 V, V_{IL} = 0.7 V, I_{OH} = -400 µA 1/	2.5		V
Low-level output voltage	V_{OL}	V_{CC} = 4.5 V, V_{IH} = 2.0 V, I_{IL} = 0.7 V, I_{OL} = 4 mA 1/		0.4	V
Input clamp voltage	V_{IC}	V_{CC} = 4.5 V, I_{IN} = -18 mA T_C = 25°C		-1.5	V
High-level input current (A < B, A > B inputs)	I_{IH1}	V_{CC} = 5.5 V, V_{IN} = 2.7 V		20	µA
High-level input current (all other inputs)	I_{IH2}	V_{CC} = 5.5 V, V_{IN} = 2.7 V		60	µA
High-level input current (A < B, A > B inputs)	I_{IH3}	V_{CC} = 5.5 V, V_{IN} = 5.5 V		100	µA
High-level input current (all other inputs)	I_{IH4}	V_{CC} = 5.5 V, V_{IN} = 5.5 V		300	µA
Low-level input current (A < B, A > B inputs)	I_{IL1}	V_{CC} = 5.5 V, V_{IN} = 0.4 V	-0.1	-.42	mA
Low-level input current (all other inputs)	I_{IL2}	V_{CC} = 5.5 V, V_{IN} = 0.4 V	-.14	-1.3	mA
Short circuit output current	I_{OS}	V_{CC} = 5.5 V 1/ 2/	-15	-130	mA
Supply current	I_{CC}	V_{CC} = 5.5 V 3/		20	mA
Propagation delay from any A or B input to A < B or A > B outputs	t_{PLH1}	C_L = 50 pF 3 gate levels 4/5/ R_L = 2 kΩ	2	42	ns
Propagation delay from any A or B input to A > B output	t_{PLH2}	C_L = 50 pF 3 gate levels 4/5/ R_L = 2 kΩ	2	42	ns
Propagation delay from A < B or A = B input to A > B output	t_{PLH3}	C_L = 50 pF 2 gate levels 4/5/ R_L = 2 kΩ	2	25	ns

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued

Test	Symbol	Conditions $-55°C \leq T_C \leq +125°C$	Limits		Unit
			Min	Max	
Propagation delay from A = B input to A = B output	t_{PLH4}	C_L = 50 pF 1 gate level 4/5/ R_L = 2 kΩ	2	28	ns
Propagation delay from A > B or A = B input to A < B output	t_{PLH5}	C_L = 50 pF 2 gate levels 4/5/ R_L = 2 kΩ	2	30	ns
Propagation delay from any A or B input to A > B or A < B output	t_{PHL1}	C_L = 50 pF 3 gate levels 4/5/ R_L = 2 kΩ	2	42	ns
Propagation delay from any A or B input to A = B	t_{PHL2}	C_L = 50 pF 3 gate levels 4/5/ R_L = 2 kΩ	2	39	ns
Propagation delay from A < B or A = B inputs to A > B output	t_{PHL3}	C_L = 50 pF 2 gate levels 4/5/ R_L = 2 kΩ	2	28	ns
Propagation delay from A = B input to A = B output	t_{PHL4}	C_L = 50 pF 1 gate level 4/5/ R_L = 2 kΩ	2	36	ns
Propagation delay from A > B or A = B input to A < B output	t_{PHL5}	C_L = 50 pF 2 gate levels 4/5/ R_L = 2 kΩ	2	26	ns

1/ All unspecified inputs grounded.

2/ Not more than one output should be shorted at a time.

3/ I_{CC} is measured with outputs open, A = B grounded, and all other inputs at 4.5 V.

4/ The longest gate level propagation delay paths are not tested (5 gate levels); however, they can be calculated: $t = tp_1$ maximum + tp_2 maximum - tp_4 minimum.

5/ Gate level references are made with respect to the logic diagram.

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3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 11 (see MIL-M-38510, appendix E).

TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III)	
	Class S devices	Class B devices
Interim electrical parameters (pre burn-in) (method 5004)	1	1
Final electrical test parameters (method 5004)	1*,2,3,7,9,10,11	1*,2,3,7,9
Group A test requirements (method 5005)	1,2,3,7,8,9,10,11	1,2,3,7,8,9,10,11
Group B test requirements (method 5005) subgroup 5	1,2,3,7,8,9,10,11	N/A
Group C end-point electrical parameters (method 5005)	N/A	1,2,3
Additional electrical subgroups for group C periodic inspections	N/A	N/A
Group D end-point electrical parameters (method 5005)	1,2,3	1,2,3

*PDA applies to subgroup 1 (see 4.2c).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Burn-in (method 1015 of MIL-STD-883).

(1) Test condition D, using the circuit shown on figure 4, or equivalent.

(2) T_A = +125°C minimum.

b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

c. The percent defective allowable (PDA) shall be as specified in MIL-M-38510.

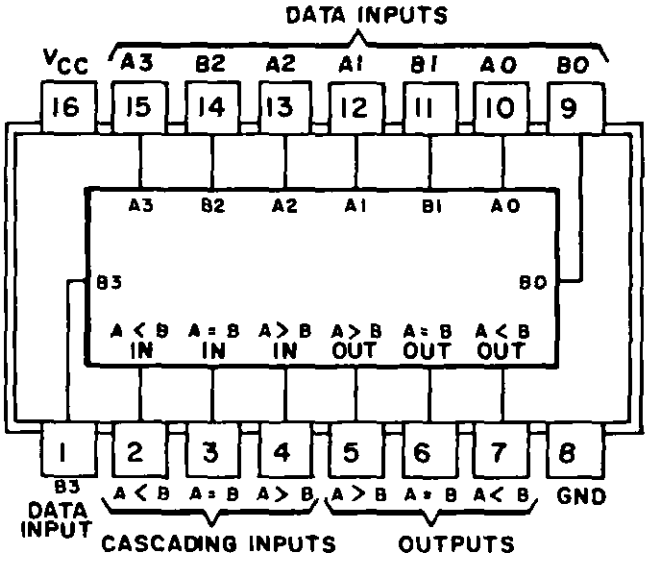
4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

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Device type 01

Cases E and F



Device type 01

Case 2

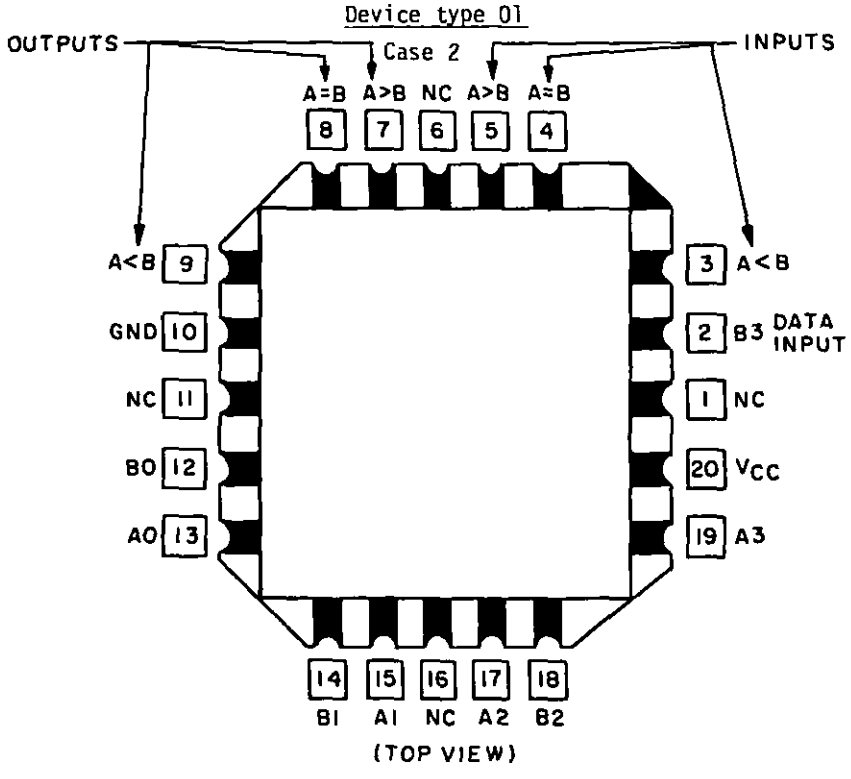


FIGURE 1. Terminal connections.

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Device type 01

COMPARING INPUTS				CASCADING INPUTS			OUTPUTS		
A3, B3	A2, B2	A1, B1	A0, B0	A > B	A < B	A = B	A > B	A < B	A = B
A3 > B3	X	X	X	X	X	X	H	L	L
A3 < B3	X	X	X	X	X	X	L	H	L
A3 = B3	A2 > B2	X	X	X	X	X	H	L	L
A3 = B3	A2 < B2	X	X	X	X	X	L	H	L
A3 = B3	A2 = B2	A1 > B1	X	X	X	X	H	L	L
A3 = B3	A2 = B2	A1 < B1	X	X	X	X	L	H	L
A3 = B3	A2 = B2	A1 = B1	A0 > B0	X	X	X	H	L	L
A3 = B3	A2 = B2	A1 = B1	A0 < B0	X	X	X	L	H	L
A3 = B3	A2 = B2	A1 = B1	A0 = B0	H	L	L	H	L	L
A3 = B3	A2 = B2	A1 = B1	A0 = B0	L	H	L	L	H	L
A3 = B3	A2 = B2	A1 = B1	A0 = B0	X	X	H	L	L	H
A3 = B3	A2 = B2	A1 = B1	A0 = B0	H	H	L	L	L	L
A3 = B3	A2 = B2	A1 = B1	A0 = B0	L	L	L	H	H	L

H = High level
L = Low level
X = Irrelevant

FIGURE 2. Truth table.

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Device type 01

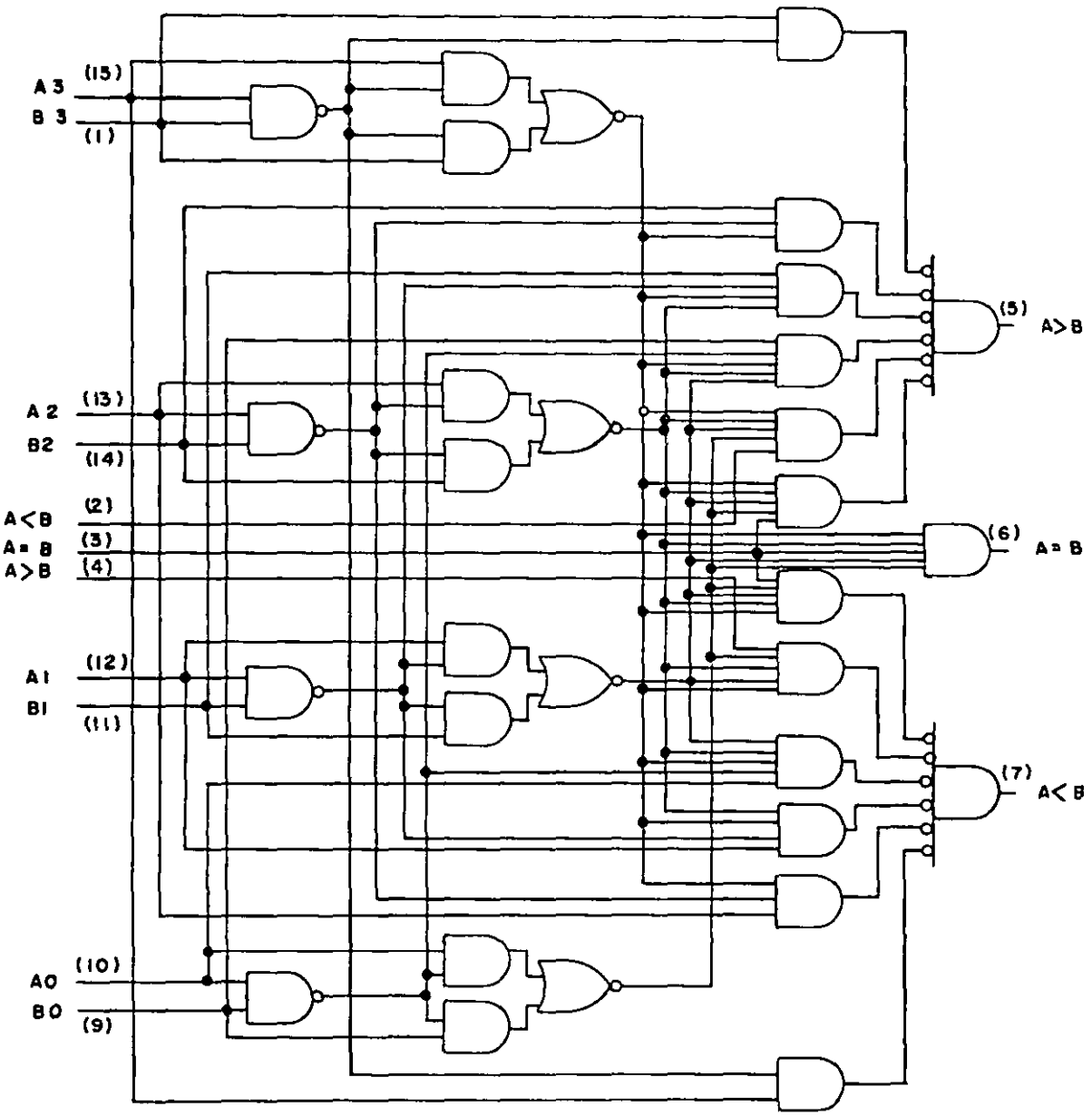
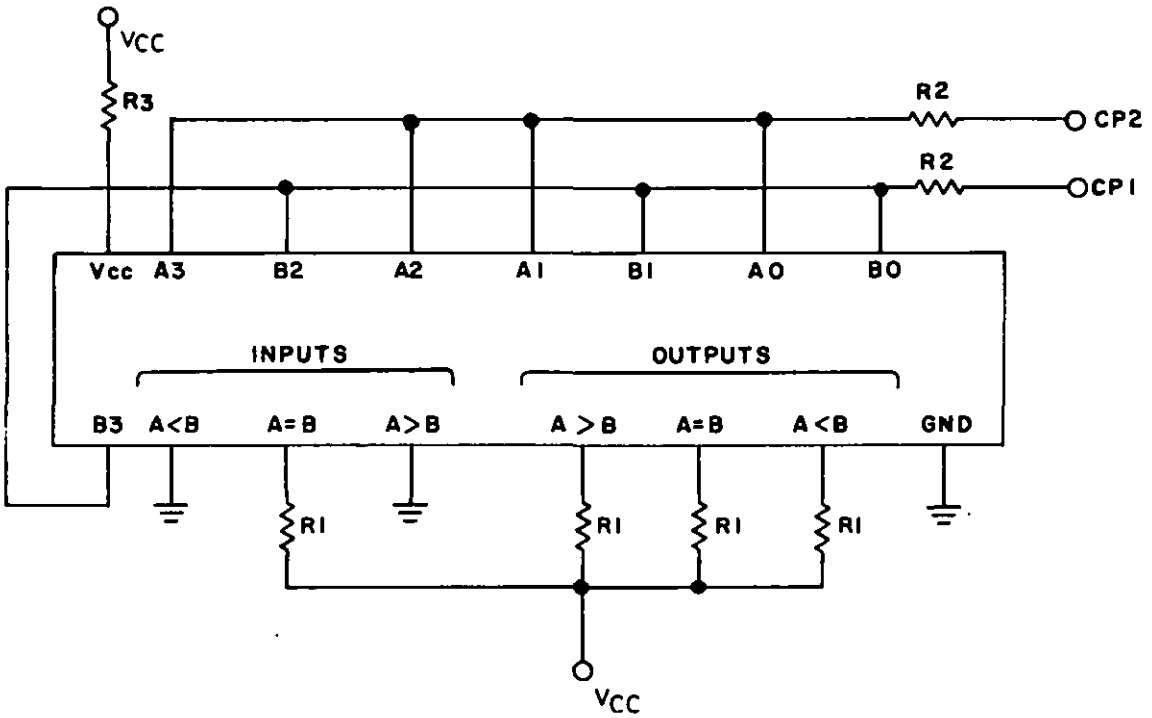


FIGURE 3. Logic diagram.

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Device type 01

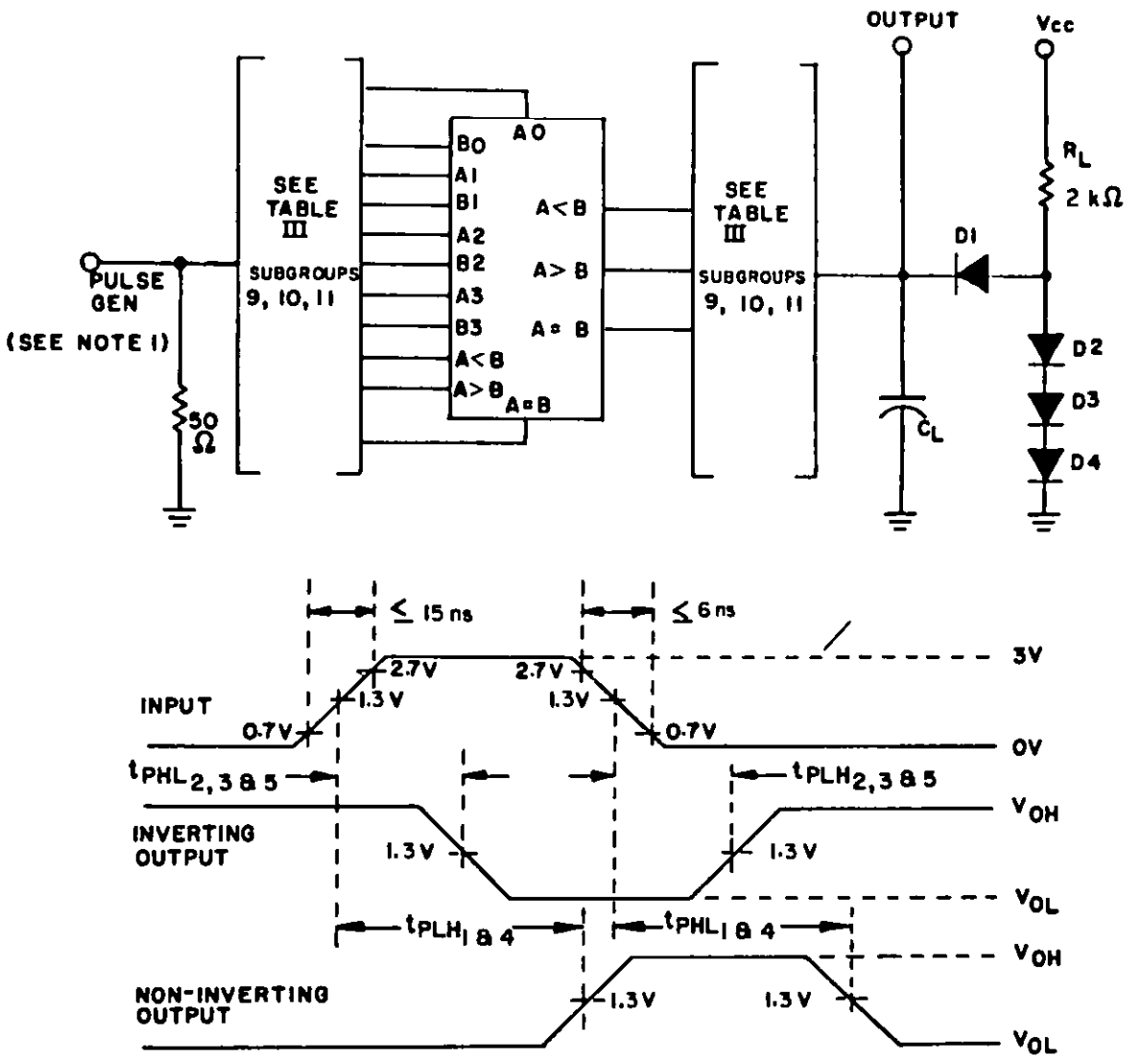


NOTES:
1. CP1 = 100 kHz ±50% square wave; duty cycle 50 ±15%; V_{IH} = 2.0 V to 5.5 V; V_{IL} = -0.5 V to 0.7 V.
2. CP2 = same as CP1 except frequency = 50 kHz ±50% square wave; CP2 synchronized with CP1.
3. R1 = 1 kΩ ±5%.
4. R2 = 27Ω ±5%.
5. R3 = 5.0Ω ±5%.
6. V_{CC} shall provide 5 V minimum at device terminal.

FIGURE 4. Burn-in and life test circuit.

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Device type 01



NOTES:
1. The pulse generator has the following characteristics: V_{gen} = 3 V, $t_1 \leq 15$ ns, $t_0 \leq 6$ ns, t_p = 500 ns, PRR $\leq$ 1 MHz, and $Z_{out} \approx 50\Omega$.
2. C_L = 50 pF and includes probe and jig capacitance.
3. All diodes are 1N3064, or equivalent.

FIGURE 5. Switching time test circuit.

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TABLE III. Group A inspection for device type 01.
Terminal conditions (pins not designated may be H ≥2.0 V, L ≤0.7 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E, F Case 1/ 2	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
				2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Min	Max	
			Test no.	B3	A < B IN	A = B IN	A > B IN	A > B OUT	A = B OUT	A < B OUT	GND	B0	A0	B1	A1	A2	B2	A3	V_{CC}				
1 T_C = +25°C	V_{OH}	3006	1	0.7 V	0.7 V	0.7 V		-400 µA			GND	0.7 V	GND	0.7 V	GND	GND	0.7 V	GND	4.5 V	A > B OUT	2.5		V
		"	2	"		2.0 V			-400 µA		"	"	0.7 V	"	0.7 V	0.7 V	"	0.7 V	"	A = B OUT	"		"
		"	3	GND		0.7 V	0.7 V			-400 µA	"	GND	"	GND	"	"	GND	"	"	A < B OUT	"		"
	V_{OL}	3007	4	2.0 V	2.0 V	GND		4 mA			"	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	"	A > B OUT		0.4	"
		"	5	"		0.7 V			4 mA		"	"	"	"	"	"	"	"	"	A = B OUT		"	"
		"	6	0.7 V		2.0 V	GND			4 mA	"	0.7 V	0.7 V	0.7 V	0.7 V	0.7 V	0.7 V	0.7 V	"	A < B OUT		"	"
	V_{IC}		7	-18 mA							"								"	B3		-1.5	"
			8		-18 mA						"								"	A < B IN		"	"
			9			-18 mA					"								"	A = B IN		"	"
			10				-18 mA				"								"	A > B IN		"	"
			11								"	-18 mA							"	B0		"	"
			12								"		-18 mA						"	A0		"	"
			13								"			-18 mA					"	B1		"	"
			14								"				-18 mA				"	A1		"	"
			15								"					-18 mA			"	A2		"	"
			16								"						-18 mA		"	B2		"	"
			17								"							-18 mA	"	A3		"	"
	I_{IH1}	3010	18	GND	2.7 V						"	GND	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	A < B IN		20	µA
		"	19	"			2.7 V				"	"	"	"	"	"	"	"	"	A > B IN		"	"
	I_{IH2}	"	20	2.7 V							"							GND	"	B3		60	"
		"	21	GND		2.7 V					"	GND	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	"	A = B IN		"	"
		"	22	"							"	2.7 V	GND	"	"	"	GND	"	"	B0		"	"
		"	23	"							"	GND	2.7 V	"	"	"	"	"	"	A0		"	"
		"	24	"							"	"	5.5 V	2.7 V	"	"	"	"	"	B1		"	"
		"	25	"							"	"	"	GND	2.7 V	"	"	"	"	A1		"	"
		"	26	"							"	"	"	"	GND	2.7 V	"	"	"	A2		"	"
		"	27	"							"	"	"	"	5.5 V	GND	2.7 V	"	"	B2		"	"
		"	28	"							"							2.7 V	"	A3		"	"
		"	29	2.7 V							"							5.5 V	"	B3		"	"
		"	30	GND							"	2.7 V	5.5 V	GND	5.5 V	5.5 V	GND	"	"	B0		"	"
		"	31	"							"	5.5 V	2.7 V	"	"	"	"	"	"	A0		"	"
		"	32	"							"	GND	5.5 V	2.7 V	GND	"	"	"	"	B1		"	"
		"	33	"							"	"	"	5.5 V	2.7 V	"	"	"	"	A1		"	"
		"	34	"							"	"	"	GND	5.5 V	2.7 V	5.5 V	"	"	A2		"	"
		"	35	"							"	"	"	"	"	5.5 V	2.7 V	"	"	B2		"	"
		"	36	5.5 V							"							2.7 V	"	A3		"	"
	I_{IH3}	"	37	GND	5.5 V						"	GND	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	"	A < B IN		100	"
		"	38	"			5.5 V				"	"	"	"	"	"	"	"	"	A > B IN		"	"
	I_{IH4}	"	39	5.5 V							"							GND	"	B3		300	"
		"	40	GND		5.5 V					"	GND	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	"	A = B IN		"	"
		"	41	"							"	5.5 V	GND	"	"	"	GND	"	"	B0		"	"
		"	42	"							"	GND	5.5 V	"	"	"	"	"	"	A0		"	"
		"	43	"							"	"	"	"	"	"	"	"	"	B1		"	"
		"	44	"							"	"	"	GND	"	"	"	"	"	A1		"	"
		"	45	"							"	"	"	"	GND	"	"	"	"	A2		"	"
		"	46	"							"	"	"	"	5.5 V	GND	5.5 V	"	"	B2		"	"
		"	47	"							"							"	"	A3		"	"
		"	48	5.5 V							"							"	"	B3		"	"
		"	49	GND							"	5.5 V	5.5 V	GND	5.5 V	5.5 V	GND	"	"	B0		"	"
		"	50	"							"	"	"	"	"	"	"	"	"	A0		"	"
		"	51	"							"	GND	"	5.5 V	GND	"	"	"	"	B1		"	"
		"	52	"							"	"	"	5.5 V	"	"	"	"	"	A1		"	"
		"	53	"							"	"	"	GND	"	"	5.5 V	"	"	A2		"	"
		"	54	"							"	"	"	"	"	"	"	"	"	B2		"	"
		"	55	5.5 V							"							"	"	A3		"	"

See footnotes at end of table.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be H ≥2.0 V, L ≤0.7 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E, F Case 1/ 2 Test no.	1 2 B3	2 3 A < B IN	3 4 A = B IN	4 5 A > B IN	5 7 A > B OUT	6 8 A = B OUT	7 9 A < B OUT	8 10 GND	9 12 B0	10 13 A0	11 14 B1	12 15 A1	13 17 A2	14 18 B2	15 19 A3	16 20 V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
1 T_C = +25°C	I_{IL1} 2/	3009	56	5.5 V	0.4 V		0.4 V				GND	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	A < B IN	-0.1	-.38	mA
		"	57	"							"	"	"	"	"	"	"	"	"	A > B IN	"	"	"
	I_{IL2} 2/3/	"	58	0.4 V							"								B3	-.40	-1.14	"
		"	59	5.5 V		0.4 V					"	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	"	A = B IN	-.32	"	"
		"	60	"							"	0.4 V	"	"	"	"	"	"	B0	-.40	"	"
		"	61	"							"	5.5 V	0.4 V	"	"	"	"	"	A0	"	"	"
		"	62	"							"	"	5.5 V	0.4 V	"	"	"	"	B1	"	"	"
		"	63	"							"	"	"	5.5 V	0.4 V	"	"	"	A1	"	"	"
		"	64	"							"	"	"	"	5.5 V	"	"	"	A2	"	"	"
		"	65	"							"	"	"	"	"	0.4 V	"	"	B2	"	"	"
		"	66	"							"	"	"	"	"	5.5 V	0.4 V	"	A3	"	"	"
	I_{OS} 4/	3011	67	GND	GND	GND		GND			"	GND	GND	GND	GND	GND	GND	GND	"	A > B OUT	-15	-100	"
		"	68	"	"	2.0 V			GND		"	"	"	"	"	"	"	"	"	A = B OUT	"	"	"
		"	69	"	2.0 V	GND	GND			GND	"	"	"	"	"	"	"	"	"	A < B OUT	"	"	"
	I_{CC}	3005	70	4.5 V	4.5 V	GND	4.5 V				"	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V	"	V_{CC}		20	"
2	Same tests, terminal conditions, and limits as subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																					
3	Same tests, terminal conditions, and limits as subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																					
7 T_C = +25°C	Truth table tests 5/	3014	71	B 6/	A 6/	B 6/	B 6/	H 6/	L 6/	L 6/	GND	A 6/	B 6/	A 6/	B 6/	B 6/	A 6/	A 6/	4.5 V	See 5/			
		"	72	A	B	"	A	L	"	H	"	B	A	B	A	A	B	B	"				
		"	73	"	A	"	B	H	"	L	"	A	B	A	B	"	"	A	"				
		"	74	"	B	"	A	L	"	H	"	B	A	B	A	B	A	"	"				
		"	75	"	A	"	B	H	"	L	"	A	B	"	"	A	"	"	"				
		"	76	"	B	"	A	L	"	H	"	B	A	A	B	"	"	"	"				
		"	77	"	A	"	B	H	"	L	"	"	"	"	A	"	"	"	"				
		"	78	"	B	"	A	L	"	H	"	A	B	"	"	"	"	"	"				
		"	79	"	"	"	"	H	"	L	"	"	A	"	"	"	"	"	"				
		"	80	"	"	A	B	L	H	"	"	"	"	"	"	"	"	"	"				
		"	81	"	"	"	A	"	"	"	"	"	"	"	"	"	"	"	"				
		"	82	"	A	B	B	"	L	H	"	"	"	"	"	"	"	"	"				
		"	83	"	"	"	A	"	"	L	"	"	"	"	"	"	"	"	"				
		"	84	"	"	A	B	"	H	"	"	"	"	"	"	"	"	"	"				
		"	85	"	"	"	A	"	"	"	"	"	"	"	"	"	"	"	"				
		"	86	"	B	B	B	H	L	H	"	"	"	"	"	"	"	"	"				
8	Repeat subgroup 7 at T_C = +125°C and -55°C.																					
9 T_C = +125°C	t_{PLH1}	3003 Fig. 5	87	IN		GND	GND			OUT	GND	GND	GND	GND	GND	GND	GND	2.7 V	5.0 V	B3 to A < B	2	35	ns
		"	88	GND		"	"			"	"	IN	2.7 V	"	"	"	"	GND	"	B0 to A < B	"	"	"
		"	89	"	GND	"		OUT			"	2.7 V	IN	"	"	"	"	"	"	A0 to A > B	"	"	"
		"	90	"		"	GND			OUT	"	GND	GND	IN	2.7 V	"	"	"	"	B1 to A < B	"	"	"
		"	91	"	GND	"		OUT			"	"	"	2.7 V	IN	"	"	"	"	A1 to A > B	"	"	"
		"	92	"	"	"	"	"			"	"	"	GND	GND	IN	2.7 V	"	"	A2 to A > B	"	"	"
		"	93	"		"	GND			OUT	"	"	"	"	"	2.7 V	IN	"	"	B2 to A < B	"	"	"
		"	94	2.7 V	GND	"		OUT			"	"	"	"	"	GND	GND	IN	"	A3 to A > B	"	"	"

See footnotes at end of table.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be H ≥2.0 V, L ≤0.7 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E, F Case 1/ 2 Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits Min	Limits Max	Unit
			2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20				
				B3	A < B IN	A = B IN	A > B IN	A > B OUT	A = B OUT	A < B OUT	GND	B0	A0	B1	A1	A2	B2	A3	V_{CC}				
9 T_C = +125°C	t_{PLH2}	3003 Fig. 5	95	IN		2.7 V			OUT		GND	GND	GND	GND	GND	GND	GND	GND	5.0 V	B3 to A = B	2	36	ns
		"	96	GND		"			"		"	IN	"	"	"	"	"	"	"	B0 to A = B	"	"	"
		"	97	"		"			"		"	GND	IN	"	"	"	"	"	"	A0 to A = B	"	"	"
		"	98	"		"			"		"	"	GND	IN	"	"	"	"	"	B1 to A = B	"	"	"
		"	99	"		"			"		"	"	"	GND	IN	"	"	"	"	A1 to A = B	"	"	"
		"	100	"		"			"		"	"	"	"	GND	IN	"	"	"	A2 to A = B	"	"	"
		"	101	"		"			"		"	"	"	"	"	GND	IN	"	"	B2 to A = B	"	"	"
		"	102	"		"			"		"	"	"	"	"	"	GND	IN	"	A3 to A = B	"	"	"
	t_{PLH3}	"	103	"	IN	GND		OUT			"	"	"	"	"	"	"	GND	"	A < B to A > B	"	20	"
		"	104	"	GND	IN		"			"	"	"	"	"	"	"	"	"	A = B to A > B	"	"	"
	t_{PLH5}	"	105	"		IN	GND			OUT	"	"	"	"	"	"	"	"	"	A = B to A < B	"	22	"
		"	106	"		GND	IN			"	"	"	"	"	"	"	"	"	"	A > B to A < B	"	"	"
	t_{PLH4}	"	107	"	IN			OUT			"	"	"	"	"	"	"	"	"	A = B to A = B	"	20	"
	t_{PHL1}	"	108	IN		GND	GND			OUT	"	"	"	"	"	"	"	2.7 V	"	B3 to A < B	"	35	"
		"	109	GND		"	"			"	"	IN	2.7 V	"	"	"	"	GND	"	B0 to A < B	"	"	"
		"	110	"	GND	"		OUT			"	2.7 V	IN	"	"	"	"	"	"	A0 to A > B	"	"	"
		"	111	"	"	"	GND			OUT	"	GND	GND	IN	2.7 V	"	"	"	"	B1 to A < B	"	"	"
		"	112	"	GND	"		OUT			"	"	"	2.7 V	IN	"	"	"	"	A1 to A > B	"	"	"
		"	113	"	"	"		"			"	"	"	GND	GND	IN	2.7 V	"	"	A2 to A > B	"	"	"
		"	114	"	"	"	GND			OUT	"	"	"	"	"	2.7 V	IN	"	"	B2 to A < B	"	"	"
		"	115	2.7 V	GND	"		OUT			"	"	"	"	"	GND	GND	IN	"	A3 to A > B	"	"	"
	t_{PHL2}	"	116	IN		2.7 V			OUT		"	"	"	"	"	"	"	GND	"	B3 to A = B	"	33	"
		"	117	GND		"			"		"	IN	"	"	"	"	"	"	"	B0 to A = B	"	"	"
		"	118	"		"			"		"	GND	IN	"	"	"	"	"	"	A0 to A = B	"	"	"
		"	119	"		"			"		"	"	GND	IN	"	"	"	"	"	B1 to A = B	"	"	"
		"	120	"		"			"		"	"	"	GND	IN	"	"	"	"	A1 to A = B	"	"	"
		"	121	"		"			"		"	"	"	"	GND	IN	"	"	"	A2 to A = B	"	"	"
		"	122	"		"			"		"	"	"	"	"	GND	IN	"	"	B2 to A = B	"	"	"
		"	123	"		"			"		"	"	"	"	"	"	GND	IN	"	A3 to A = B	"	"	"
	t_{PHL3}	"	124	"	IN	GND		OUT			"	"	"	"	"	"	"	GND	"	A < B to A > B	"	20	"
		"	125	"	GND	IN		"			"	"	"	"	"	"	"	"	"	A = B to A > B	"	"	"
	t_{PHL5}	"	126	"		IN	GND			OUT	"	"	"	"	"	"	"	"	"	A = B to A < B	"	17	"
		"	127	"		GND	IN			"	"	"	"	"	"	"	"	"	"	A > B to A < B	"	"	"
	t_{PHL4}	"	128	"		IN			OUT		"	"	"	"	"	"	"	"	"	A = B to A = B	"	25	"
10	Same tests and terminal conditions as subgroup 9, except T_C = +125°C, and test limits as follows; in minimum limits column limits for all tests are 2 ns; in maximum limits column, limits for t_{PLH1} and t_{PLH2} are 42 ns, t_{PLH3} are 25 ns, t_{PLH5} are 30 ns, t_{PLH4} are 28 ns, t_{PHL1} are 42 ns, t_{PHL2} are 39 ns, t_{PHL3} are 28 ns, t_{PHL5} are 26 ns, t_{PHL4} are 36 ns.																						
11	Same tests, terminal conditions, and limits as subgroup 10, except T_C = -55°C.																						

1/ Case 2 pins not referenced are NC.
2/ For circuit F, I_{IL1} limits for tests 56 and 57 are 0.5 to 400 µA. I_{IL2} limits for pin 3 are 0.14 to 1.14 mA, and I_{IL2} limits for remaining pins are 0.28 to 1.14 mA.
For circuit B: I_{IL1} limits for tests 56 and 57 are -180/-420 µA. I_{IL2} limits for tests 58 thru 66 are -400/-1300 µA.
3/ For circuit C, I_{IL2} maximum limits are 1.2 mA.
4/ For circuit B, I_{OS} limits are -30 to -130 mA.
5/ Tests shall be performed in sequence.
6/ A = 3.0 V min; B = 0.0 V or GND; H >1.5 V; L < 1.5 V.

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4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Electrical test requirements shall be as specified in table II herein.

b. Subgroups 4, 5, and 6 of table I of method 5005 of MIL-STD-883 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883. Electrical test requirements shall be as specified in table II herein.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. Steady-state life test (method 1005 of MIL-STD-883) conditions:

(1) Test condition D, using the circuit shown on figure 4, or equivalent.

(2) T_A = +125°C minimum.

(3) Test duration: 1,000 hours, except as permitted by appendix B of MIL-M-38510 and method 1005 of MIL-STD-883.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be specified as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the referenced terminal.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logisitic support of existing equipment.

6.2 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action and reporting of results, if applicable.

f. Requirements for product assurance options.

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g. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements shall not apply to direct purchase by or direct shipment to the Government.

h. Requirements for "JAN" marking.

6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

GND - - - - - - - - - - - - - - - Ground zero voltage potential.
I_{IN} - - - - - - - - - - - - - - Current flowing into an input terminal.
V_{IN} - - - - - - - - - - - - - - Voltage level at an input terminal.

6.4 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.5 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	54LS85

6.6 Manufacturers' designators. Manufacturers' circuits which form a part of this specification are designated with an "X" as shown in table IV herein.

TABLE IV. Manufacturers' designations.

Device type	Circuits					
	A	B	C	D	E	F
	Texas Instruments	National Semiconductor Corp.	Signetics Corp.	Motorola Inc.	Raytheon Co.	Fairchild Semiconductor
01	X	X	X	X	X	X

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 17

Preparing activity:
Air Force - 17

(Project 5962-0652)

Review activities:
Army - AR, MI
Navy - OS, SH
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Agent:
DLA - ES

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-38510/311B	MICROCIRCUITS, DIGITAL, LOW-POWER SCHOTTKY TTL, MAGNITUDE COMPARATORS, MONOLITHIC SILICON

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION (YYMMDD)

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